



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

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Recap

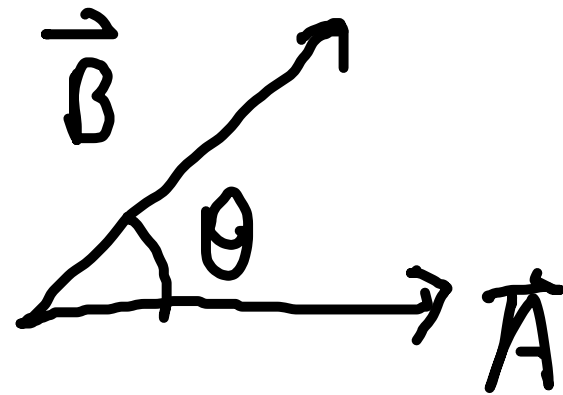
- Measurements
 - Only use SI units in PHY1001
 - Be mindful of uncertainties and significant figures
 - Estimate the order of magnitude
- Vector
 - A quantity with both magnitude and direction
 - Vector calculation: addition, scalar product, and vector product

Multiplying Vectors: Vector Product

- Yields another vector
- Also called cross product

$$\vec{C} = \vec{A} \times \vec{B}$$

Method I :



$$|\vec{C}| = |\vec{A}| |\vec{B}| \sin \theta$$

direction: right hand rule

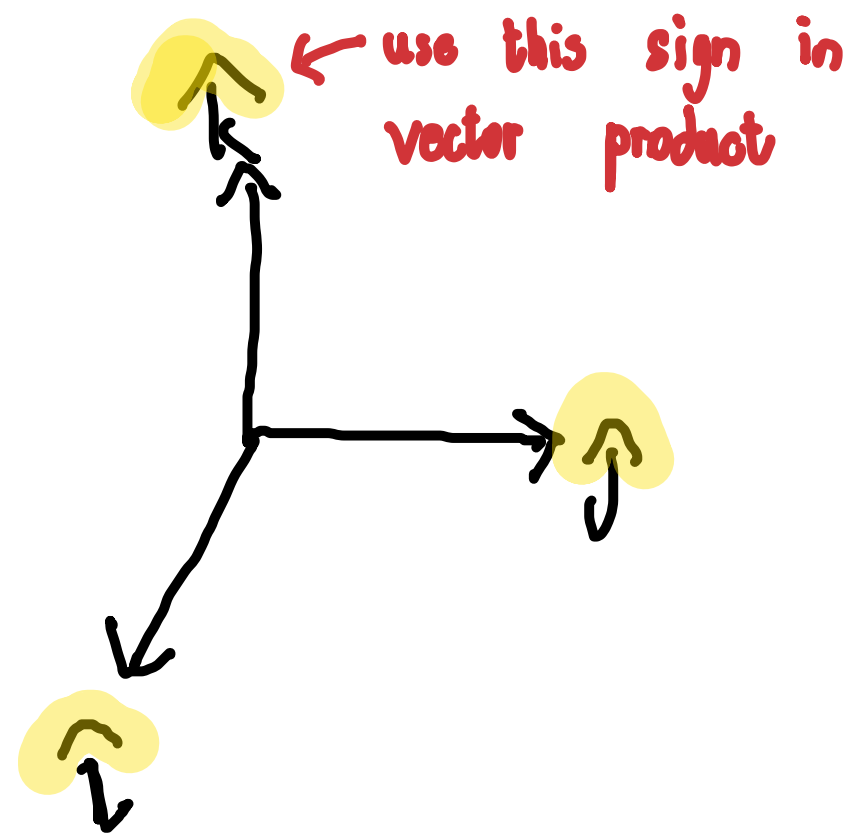
⊙ out of the page

⊗ into the page

$$\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$$

Multiplying Vectors: Vector Product

Method II : unit vector



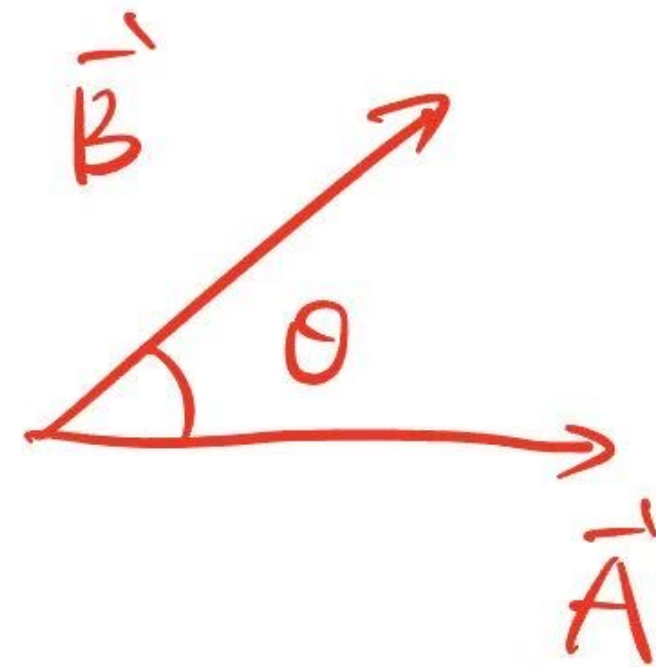
$$\hat{i} \times \hat{i} = 0 = \hat{j} \times \hat{j} = \hat{k} \times \hat{k}$$

$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}$$

$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (\quad)$$

$$= (A_x B_y - A_y B_x) \hat{k} + (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - B_z A_x) \hat{j}$$

Summary: Multiplying Vectors



- Scalar product

↳ must x hv direction

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

- Vector product

↙ direction

$$|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

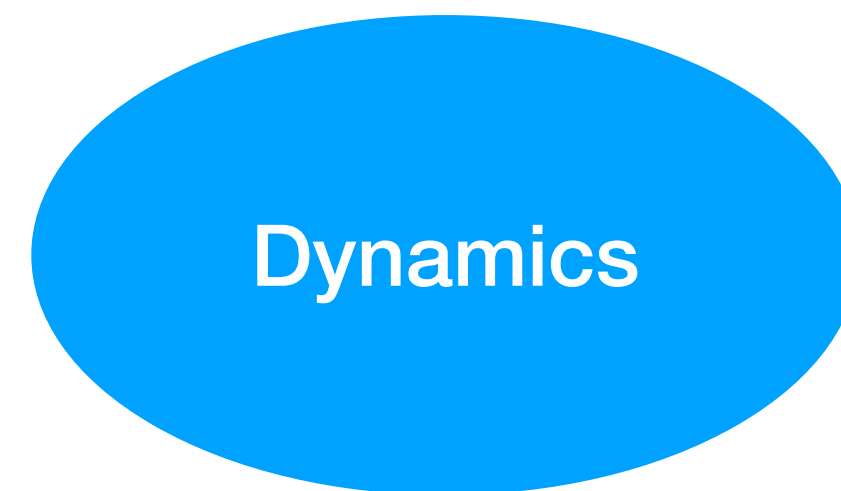
Motion

Outline

- 1D motion
- 2,3-D motion



how to move



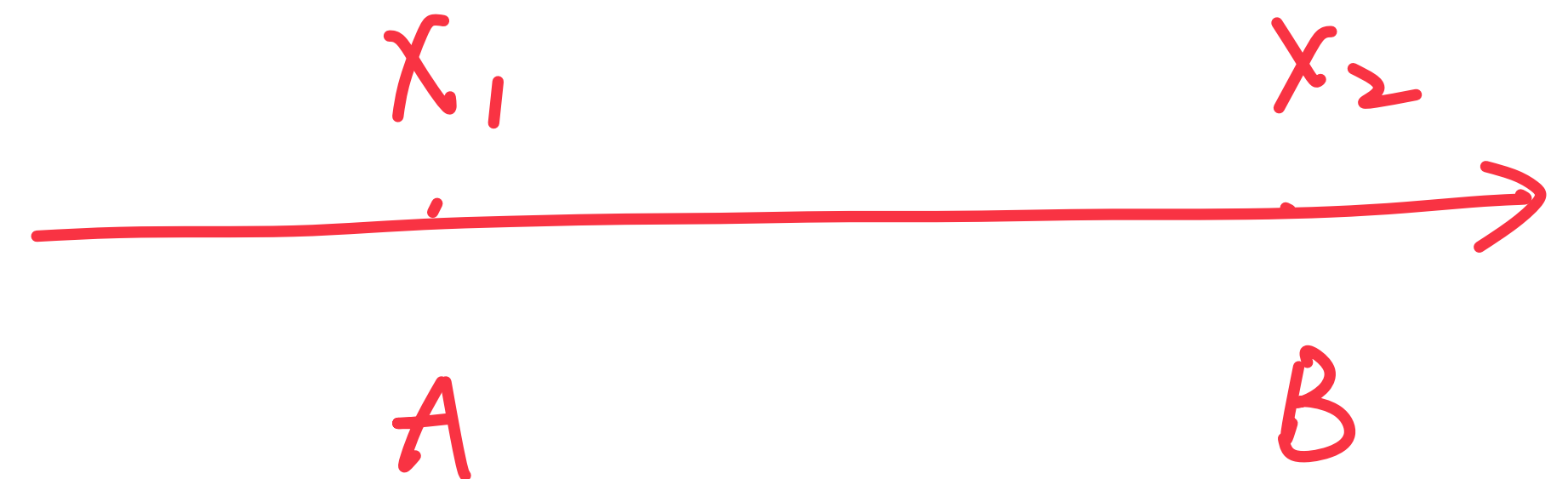
how to move
why to move

$$F = ma$$

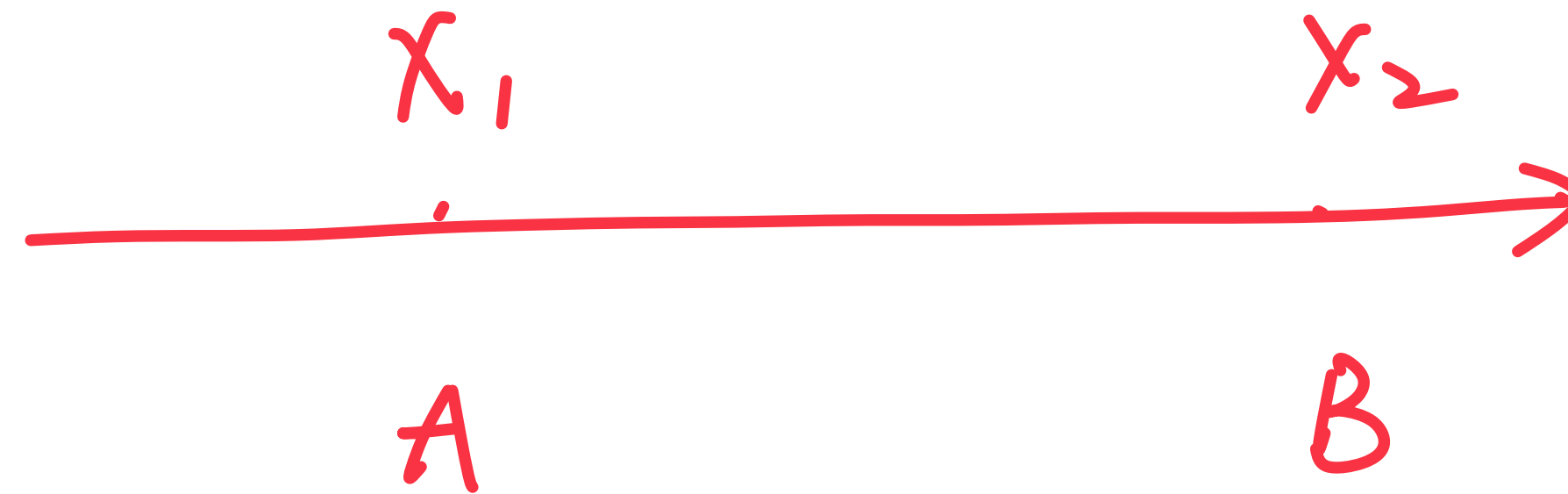
Displacement and Distance

- Displacement: Change of positions, depending on the initial and final positions, vector, denoted by x .
- Distance: depends on the path, scalar, denoted by s .

- For 1D motion, you need to specify the positive direction!
- It is your free choice, either to the left or to the right, but must pick one.

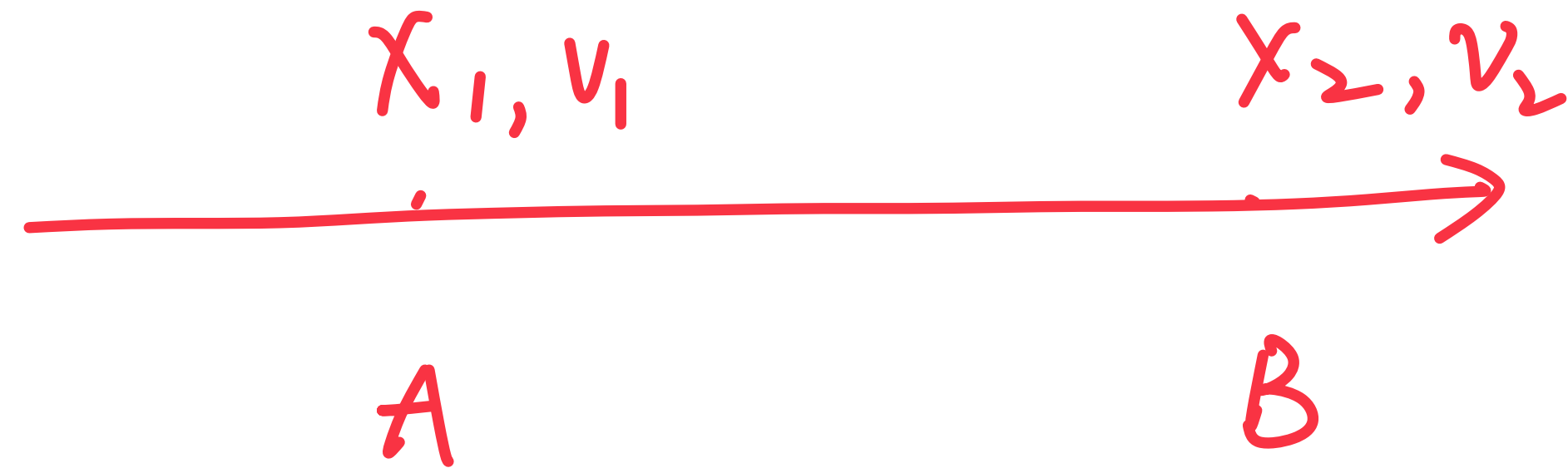


Average Velocity and Speed



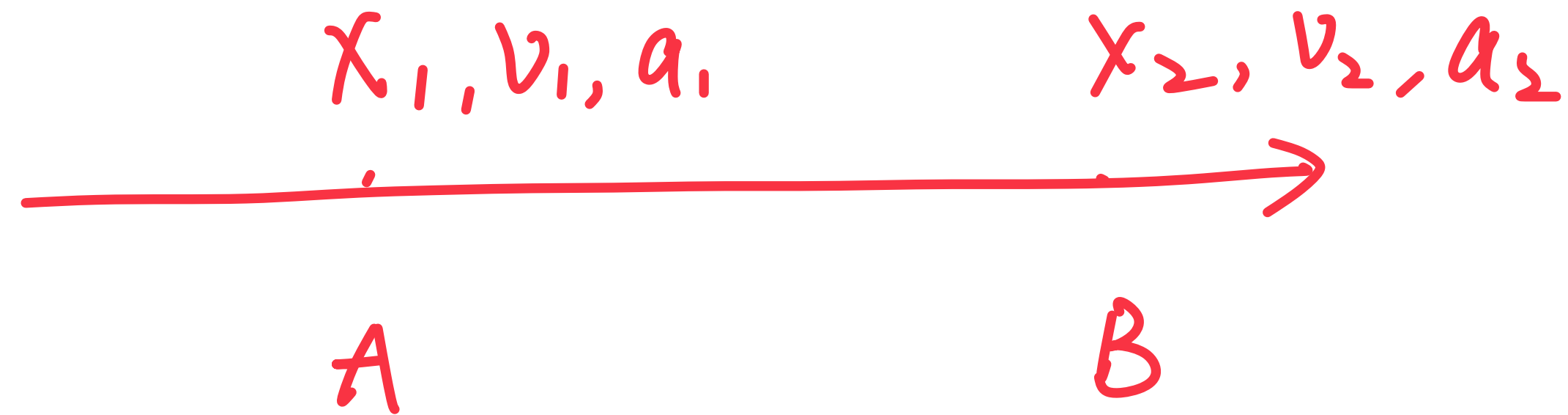
- Average velocity $v_{avg} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$, vector
- Average speed = $\frac{\Delta s}{\Delta t}$, scalar
← distance (you hv walk $\odot$)

Instantaneous Velocity and Speed



- Instantaneous velocity $\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$, vector
 - Instantaneous speed $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{|\Delta x|}{\Delta t} = |\vec{v}|$, scalar
- don't hv time to change the direction

Acceleration



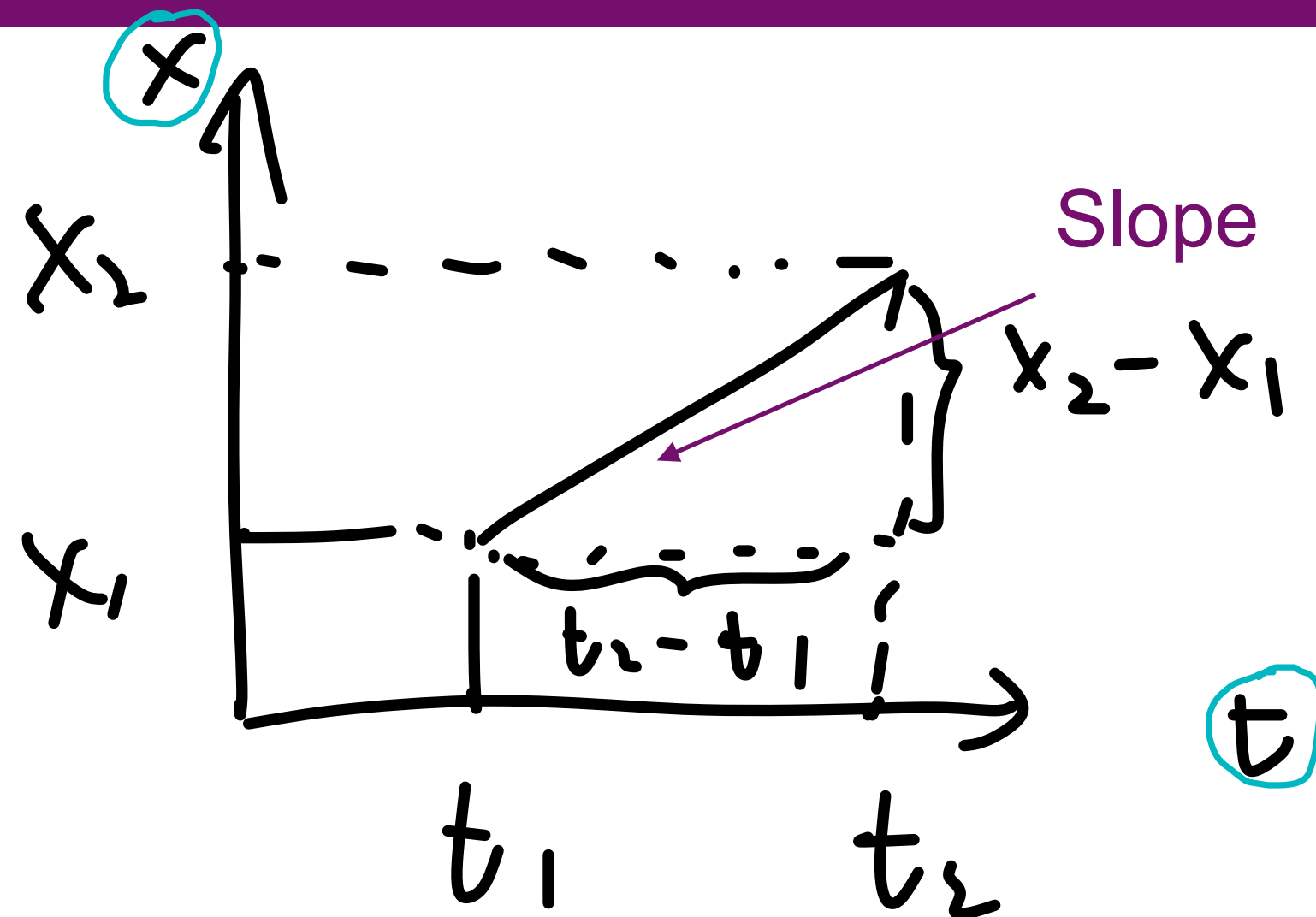
Average acceleration $a_{avg} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$

Instantaneous acceleration $a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$

In addition, we have $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

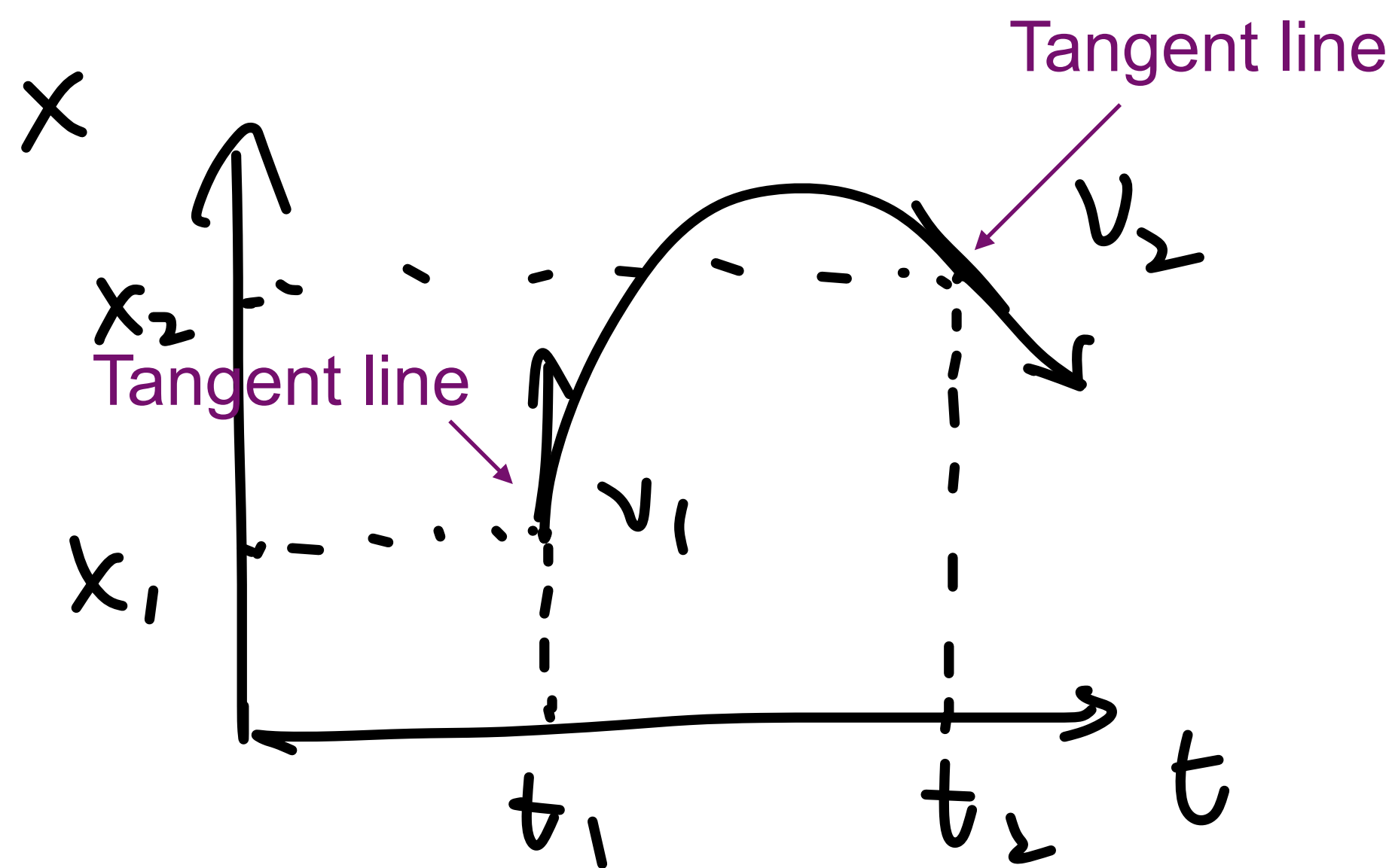
A green arrow points from the expression $v = \frac{dx}{dt}$ to the dv in the numerator of the first fraction in the equation above.

X-t Graph



$$\text{Average velocity } v_{avg} = \frac{x_2 - x_1}{t_2 - t_1}$$

velocity can be -ve
(opp to the positive direction you choose)



$$(\text{Instantaneous}) \text{ velocity } v = \frac{dx}{dt}$$

1D Motion with a Constant Acceleration

$$a = \frac{dv}{dt} \rightarrow v = at + v_0$$

$$A: (t_0, v_0, a)$$

$$B: (t_1, v_1, a)$$

$$a dt = dv$$

$$\int_{t_0}^{t_1} a dt = \int_{v_0}^{v_1} dv$$

$$a(t_1 - t_0) = v_1 - v_0$$

$$\text{Let } t_0 = 0, v_1 = v, t_1 = t$$

$$at = v - v_0 \rightarrow \boxed{v = at + v_0}$$

1D Motion with a Constant Acceleration

$$v = at + v_0 \quad v = \frac{dx}{dt} \rightarrow x = x_0 + \frac{1}{2}at^2 + v_0t$$

$$\int_0^t v dt = \int_{x_0}^x dx$$

$$\int_0^t (at + v_0) dt = x \Big|_{x_0}^x = x - x_0$$

$$\frac{1}{2}at^2 + v_0t \Big|_0^t = x - x_0$$

$$\frac{1}{2}at^2 + v_0t = x - x_0$$

$$x = \frac{1}{2}at^2 + v_0t + x_0$$

$$A (t=0, v_0, a, x_0)$$

$$B (t, v, a, x)$$

1D Motion with a Constant Acceleration

$$\underline{v = at + v_0} \quad \underline{x = x_0 + \frac{1}{2}at^2 + v_0t}$$

Get rid of t

$$2a(x - x_0) = v^2 - v_0^2$$

$$t = \frac{v - v_0}{a}$$

$$\begin{aligned} x &= x_0 + \frac{1}{2}a \left(\frac{v - v_0}{a} \right)^2 + v_0 \frac{v - v_0}{a} \\ &= x_0 + \frac{1}{2} \frac{(v - v_0)^2}{a} + \frac{2(v_0 v - v_0^2)}{2a} \\ &= x_0 + \frac{v^2 - 2vv_0 + v_0^2 + 2v_0 v - 2v_0^2}{2a} \\ &= x_0 + \frac{v^2 - v_0^2}{2a} \end{aligned}$$

$$x - x_0 = \frac{v^2 - v_0^2}{2a}$$

$$\boxed{2a(x - x_0) = v^2 - v_0^2}$$

1D Motion with a Constant Acceleration

$$v = at + v_0 \quad x = x_0 + \frac{1}{2}at^2 + v_0t \quad \xrightarrow{\text{Get rid of } a} \quad x = x_0 + \frac{1}{2}(v + v_0)t$$

$$a = \frac{v - v_0}{t}$$

$$x = x_0 + \frac{1}{2} \frac{v - v_0}{t} t^2 + v_0 t$$

$$= \underline{x_0} + \frac{v + v_0}{2} \underline{t}$$

↑
avg velocity

Four Master Equations

$$v = at + v_0$$

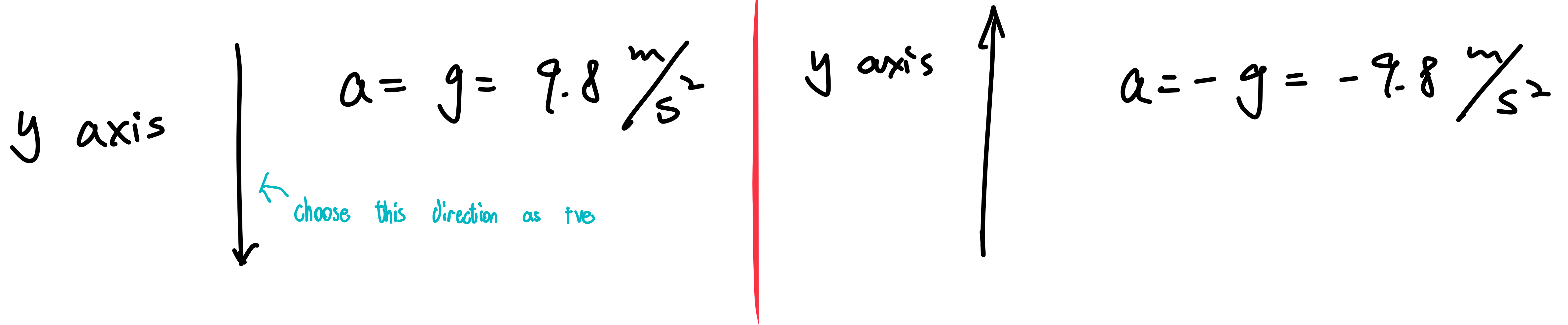
$$x = x_0 + \frac{1}{2}at^2 + v_0t$$

$$2a(x - x_0) = v^2 - v_0^2$$

$$x = x_0 + \frac{1}{2}(v + v_0)t$$

Free Fall

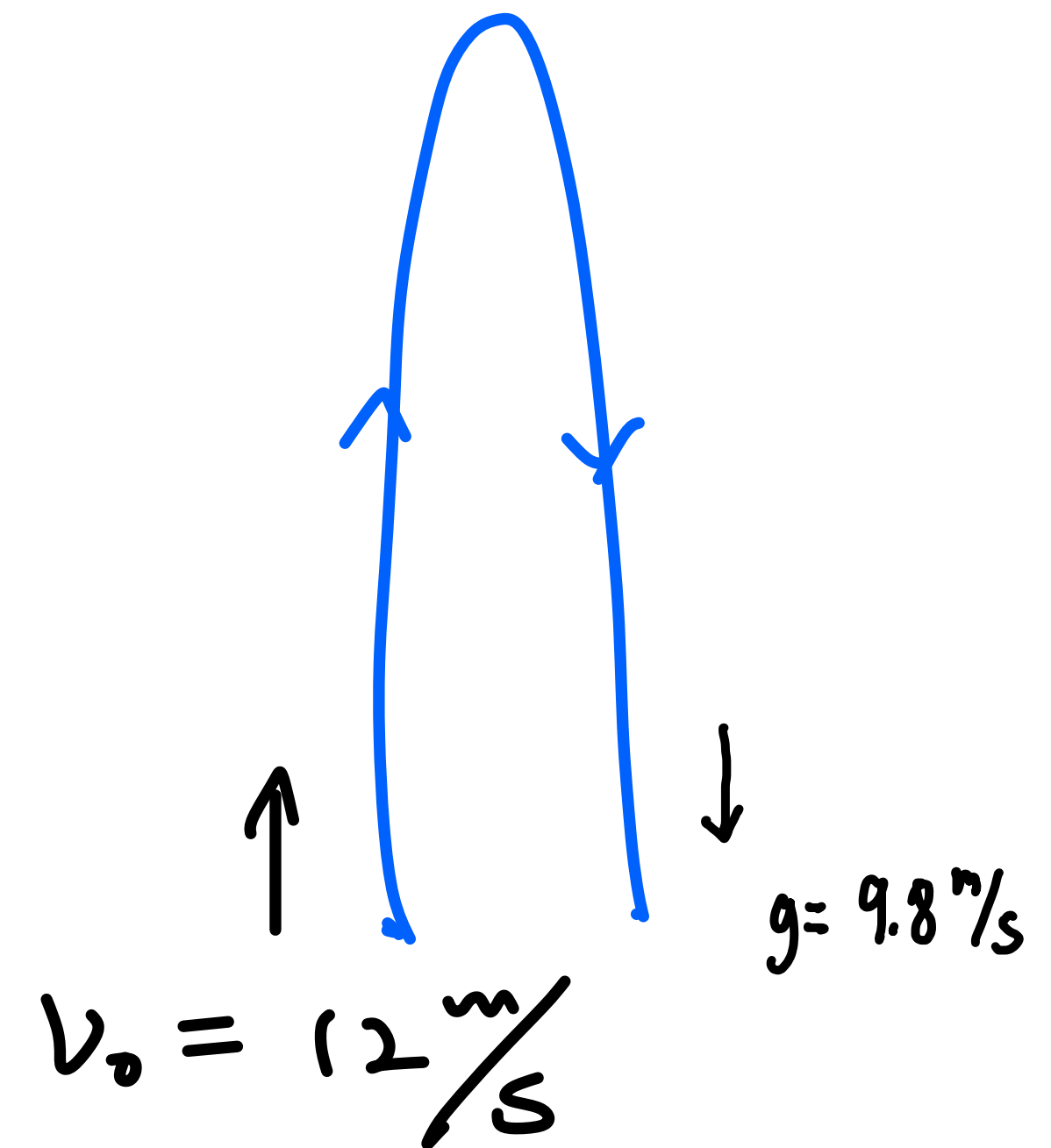
- Free fall is a motion that is only subject to gravity
- It is a motion with a constant acceleration!
- Gravitational acceleration $g = 9.8 \text{ m/s}^2$, always downwards!



Exercise

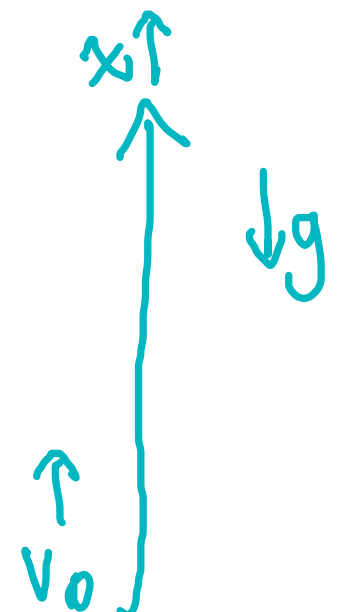
A ball is thrown upwards with an initial speed of 12 m/s , $\text{gravity} = 9.8 \text{ m/s}^2$

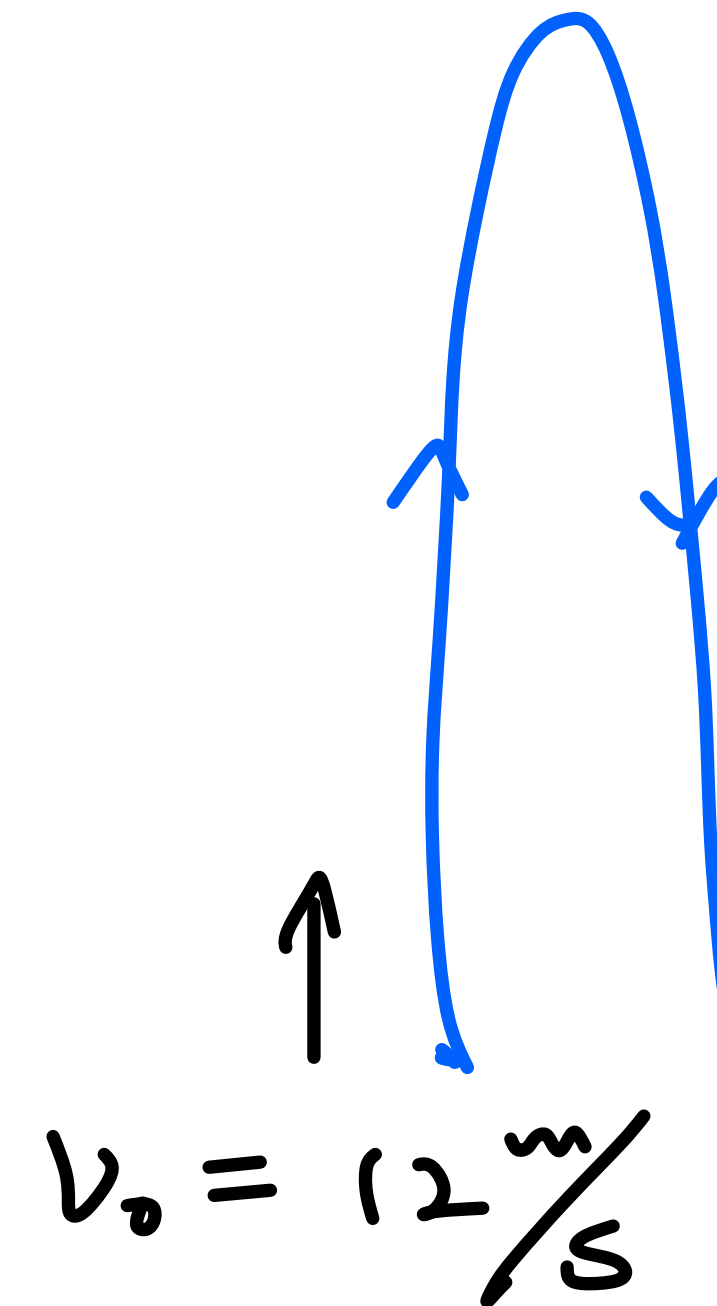
- a) How long does it take to reach its maximum height?
- b) What is the maximum height above the initial position?
- c) How long does the ball take to reach a point 5.0 m above the initial position?



Exercise

a) How long does it take to reach its maximum height?

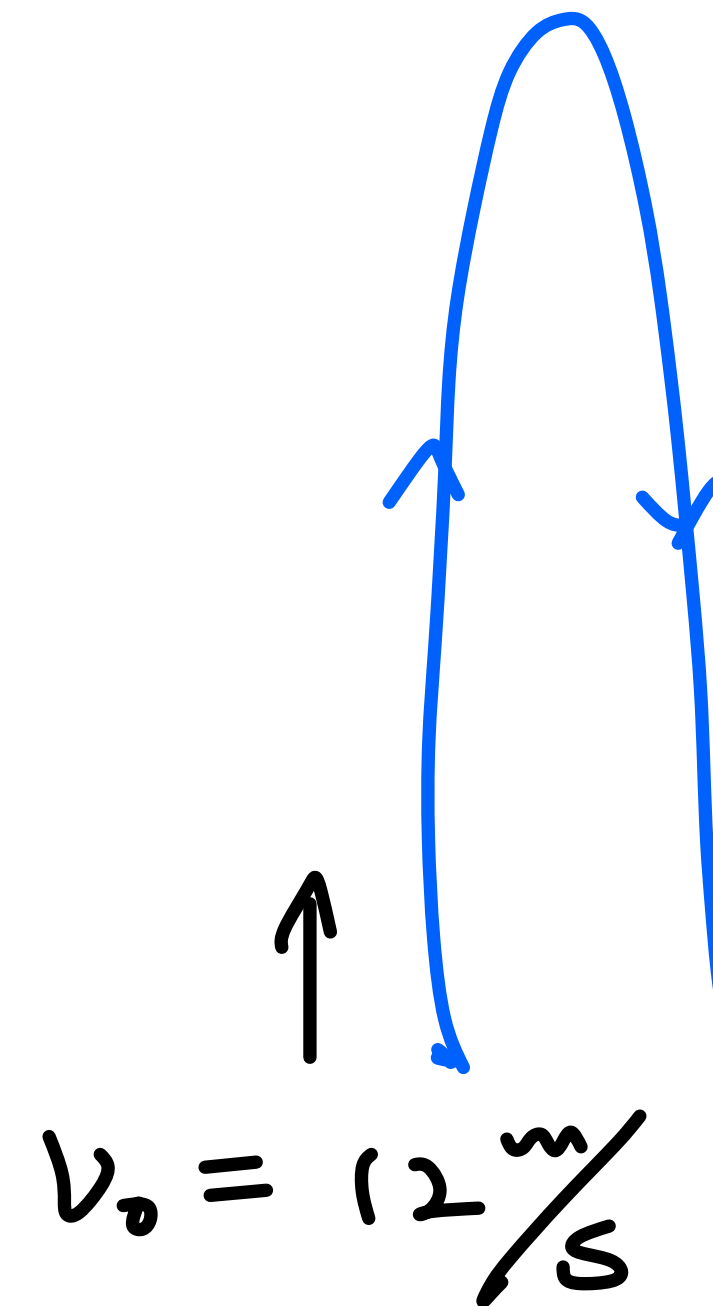

$$\begin{aligned}v_0 &= +12 \text{ m/s} \\a &= -g = -9.8 \text{ m/s}^2 \\v &= 0 \\v &= v_0 + at \\t &= \frac{v - v_0}{a} = \frac{0 - 12}{-9.8} \text{ s} \\&= 1.2 \text{ s} \quad \leftarrow 2 \text{ sig. dig.}\end{aligned}$$



Exercise

b) What is the maximum height above the initial position?

$$\begin{aligned}2a(x - x_0) &= v^2 - v_0^2 \\x - x_0 &= \frac{v^2 - v_0^2}{2a} \\&= \frac{0 - 144}{-2 \cdot 9.8} \text{ m} \\&= +7.3 \text{ m}\end{aligned}$$



Exercise

c) How long does the ball take to reach a point 5.0 m above the initial position?

$$x = +5.0 \text{ m}$$

$$x - x_0 = \frac{1}{2}at^2 + v_0t$$

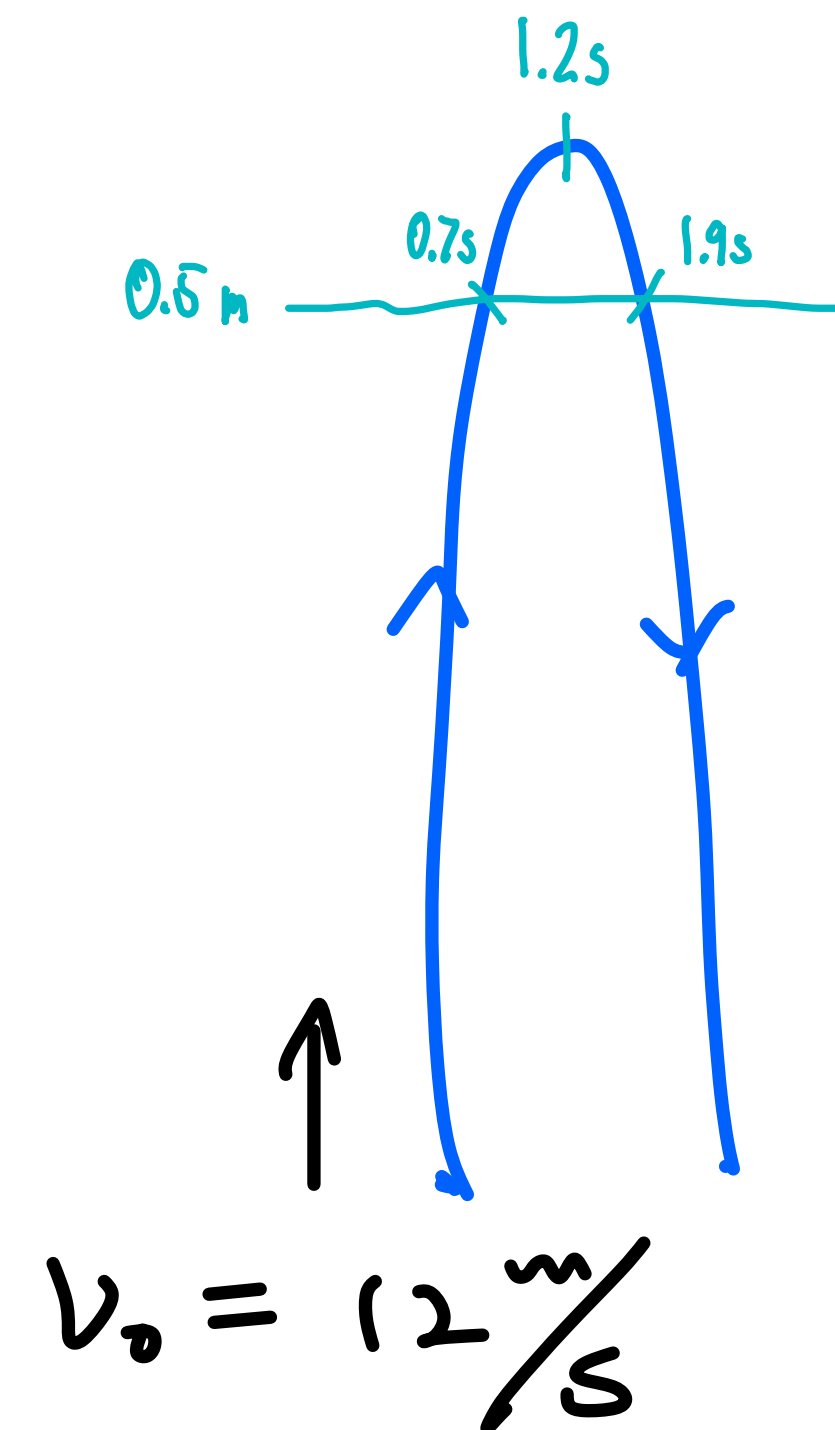
$$\frac{1}{2}at^2 + v_0t - (x - x_0) = 0$$

$$-4.9t^2 + 12t - 5 = 0$$

$$t = \frac{-12 \pm \sqrt{12^2 - 4 \times 5 \times 49}}{-2 \cdot 49} \text{ s}$$

$$= 0.53 \text{ s} \quad \text{or} \quad 1.9 \text{ s}$$

↑ ↻



3D Motion

Generalize 1D motion to higher dimensions!

$$x \longrightarrow \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$
$$= v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

Motion in each direction is **independent**!

2D Motion

Basically reduce $\hat{k}$ from the 3D Motion...

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\begin{aligned}\vec{v} &= \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} \\ &= v_x\hat{i} + v_y\hat{j}\end{aligned}$$

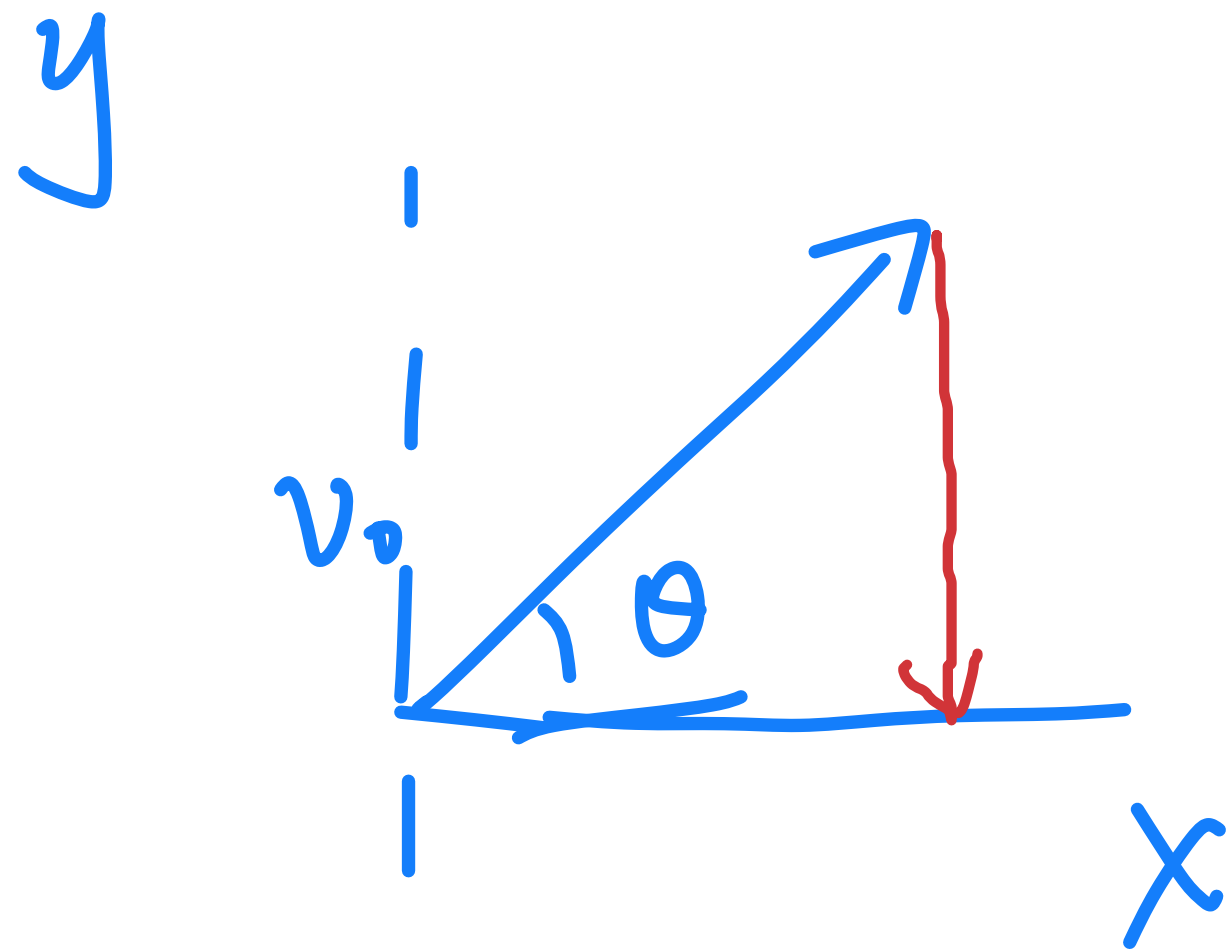
$$\vec{a} = \frac{d\vec{v}}{dt} = a_x\hat{i} + a_y\hat{j}$$

Two interesting examples:

- Projectile motion
- Uniform circular motion

Projectile Motion

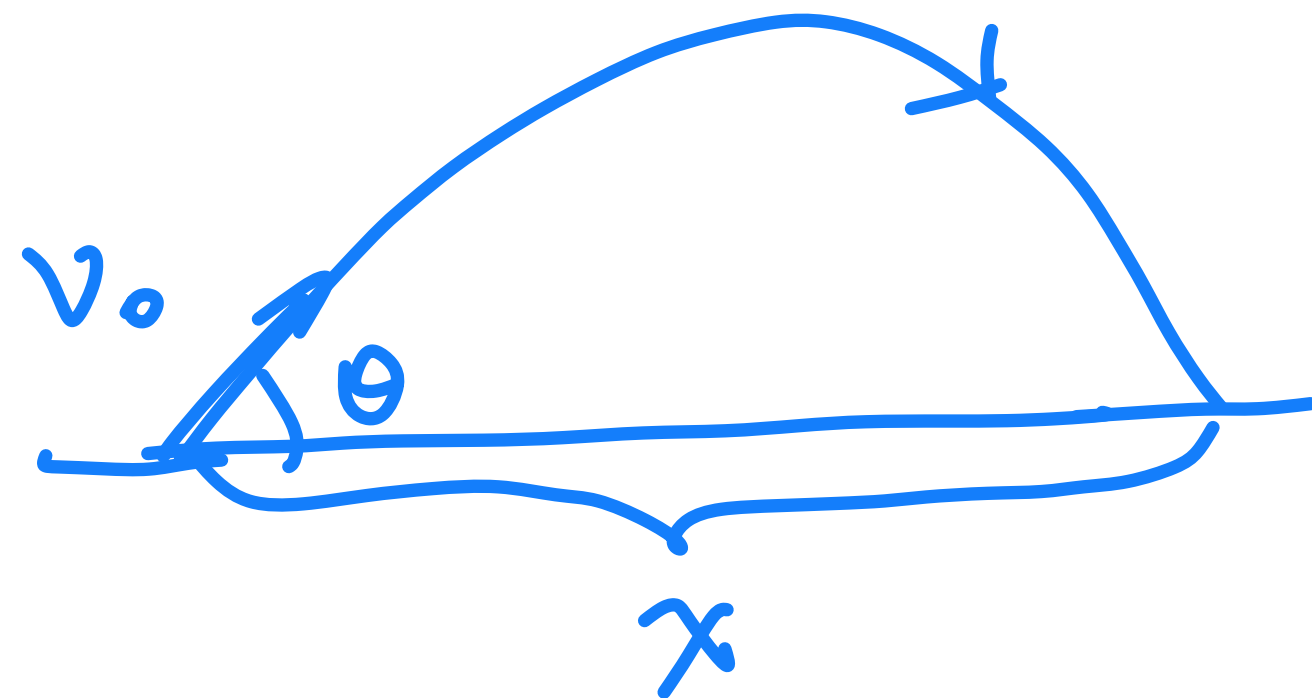
- Only subject to gravity
- With horizontal speed



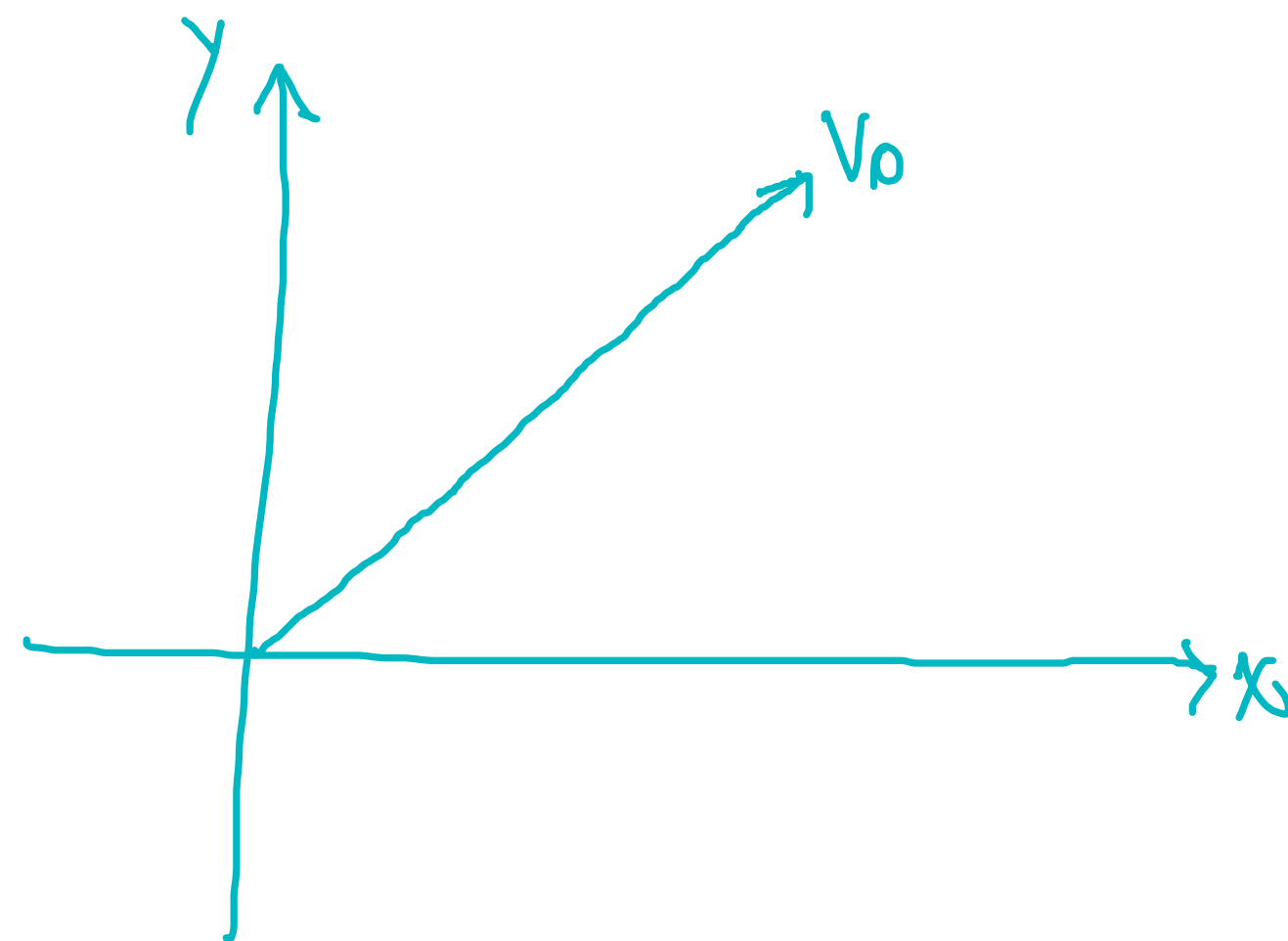
$$\underline{v_x = v_0 \cos \theta}$$

$$v_y = v_0 \sin \theta$$

Projectile Motion



A ball is thrown with a speed of v_0 at an angle θ with respect to horizontal plane. What is the horizontal distance x when the ball hits the horizontal ground again?



y:

$$\uparrow v_0 \sin \theta$$

$$g = 9.8 \text{ m/s}^2$$

$$a = -g$$

$$V = -v_0 \sin \theta$$

$$t = \frac{V - v_0}{a} = \frac{-v_0 \sin \theta - v_0 \sin \theta}{-g}$$

$$= \frac{2v_0 \sin \theta}{g}$$

x:

$$v_x = v_0 \cos \theta$$

$$x = vt = v_0 \cos \theta \cdot \frac{2v_0 \sin \theta}{g}$$

$$= \frac{v_0^2 2 \sin \theta \cos \theta}{g}$$

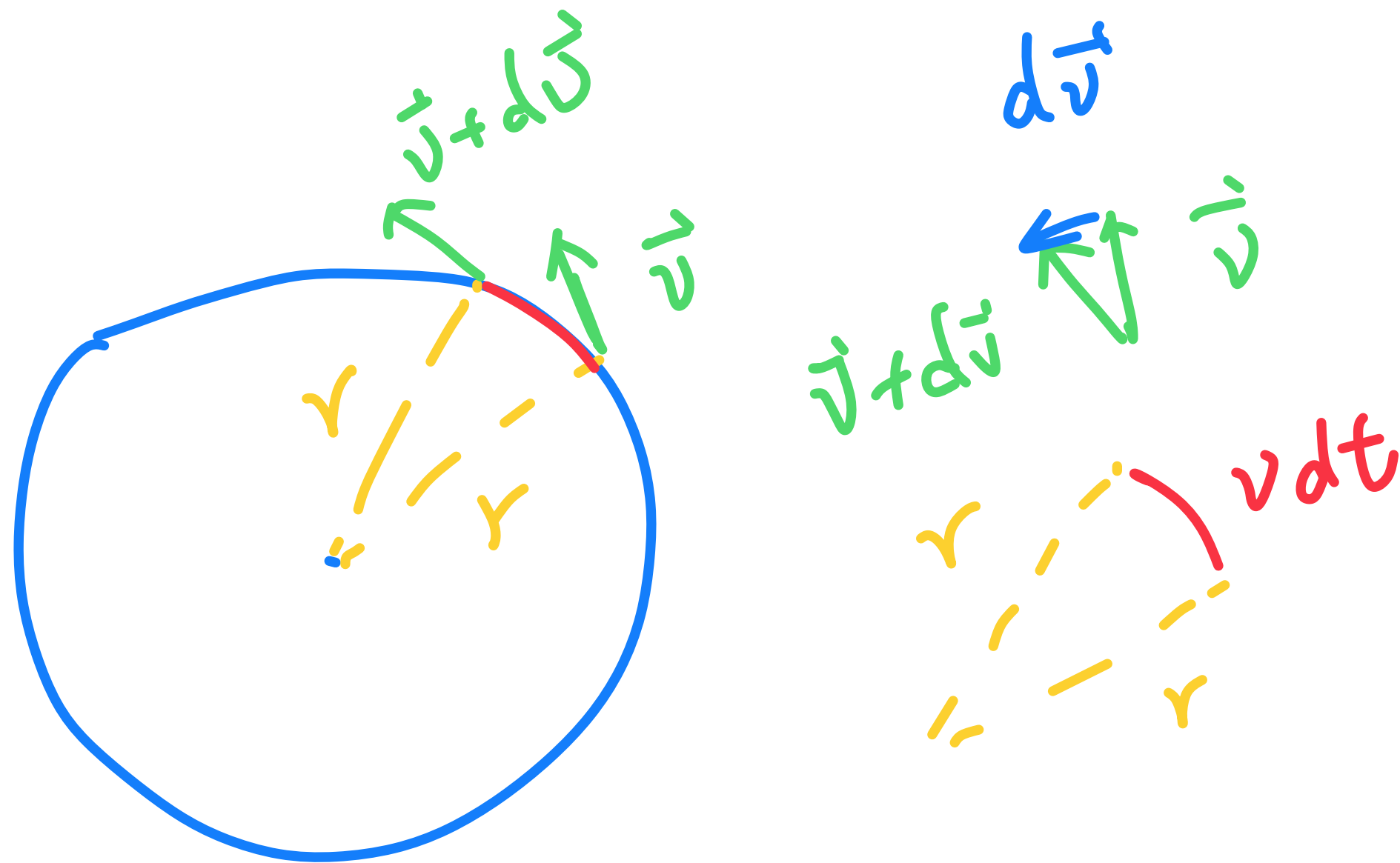
$$2 \sin \theta \cos \theta = \sin 2\theta$$

$$x = \frac{v_0^2 \sin 2\theta}{g}$$

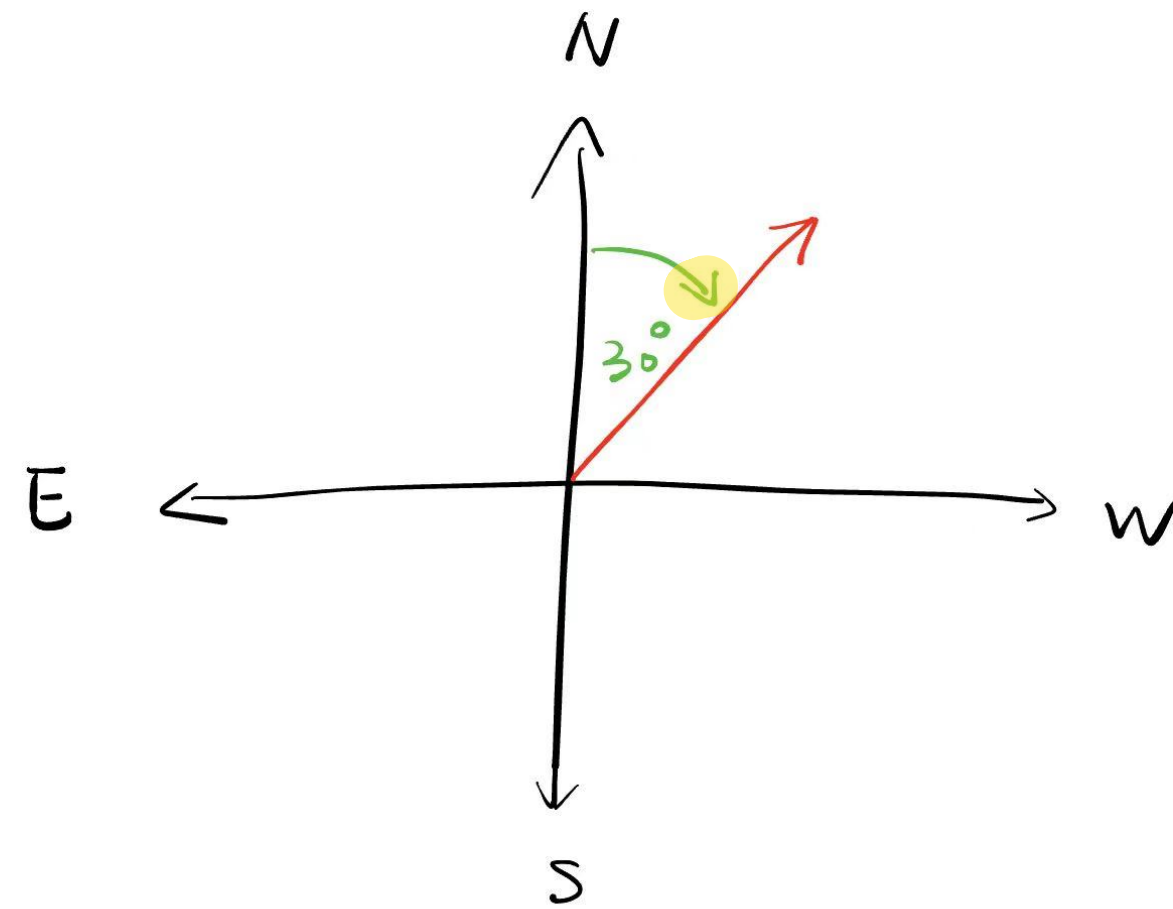
Uniform Circular Motion

2D motion in a **circular** trajectory with a **constant speed**.

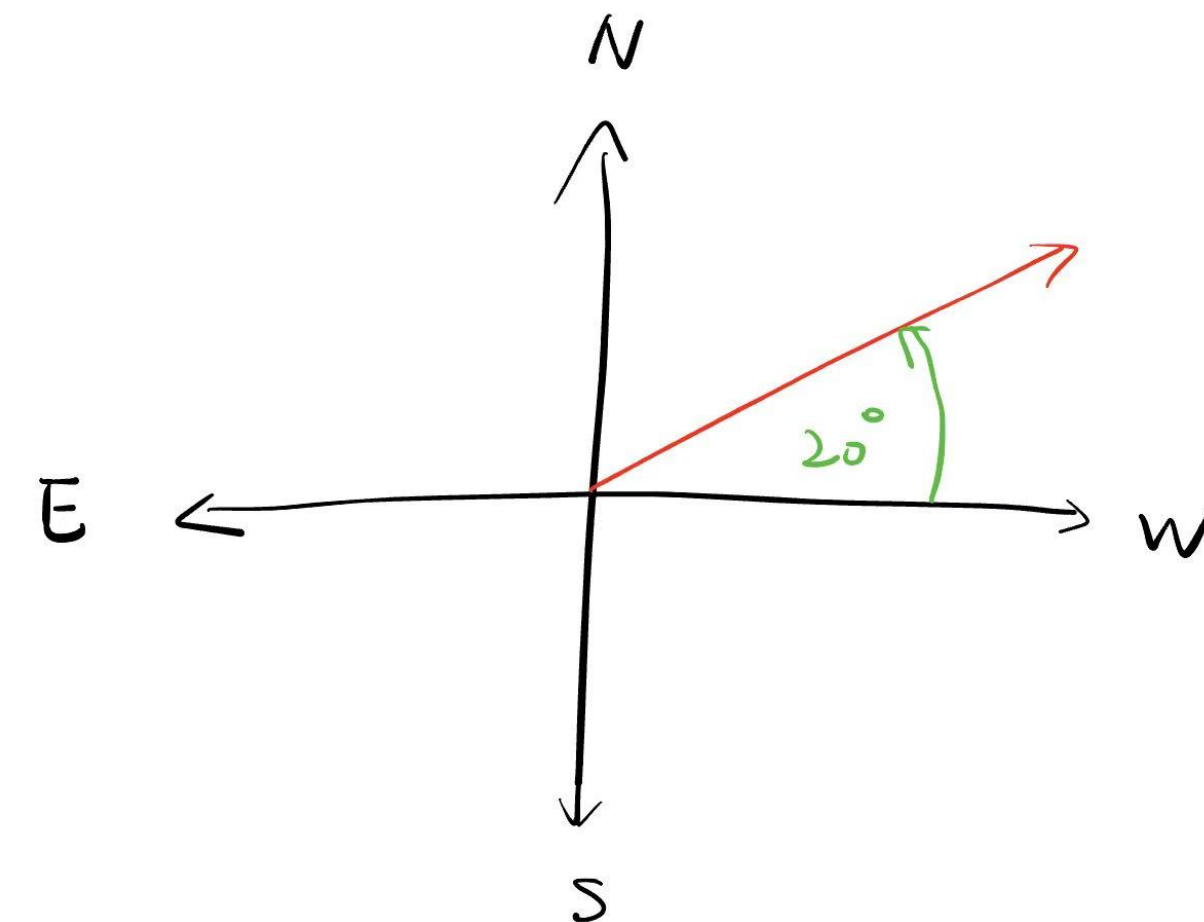
$$a = \frac{v^2}{r} = \frac{4\pi^2}{T^2}r = \omega^2 r$$



Direction in 2D Motion



30 degree west of due north
east?



20 degree north of due west

Summary: Motion

- 1D motion $v = \frac{dx}{dt}$ $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$
 - 1D motion with a constant acceleration $v = at + v_0$ $2a(x - x_0) = v^2 - v_0^2$
 $x = x_0 + \frac{1}{2}at^2 + v_0t$ $x = x_0 + \frac{1}{2}(v + v_0)t$
 - Free fall: $|a| = g = 9.8 \text{ m/s}^2$
- 2D motion
 - Projectile motion
 - Uniform circular motion $a = \frac{v^2}{r} = \frac{4\pi^2}{T^2}r = \omega^2r$
- 3D motion